

Quantum Mechanics – Homework 3

Name: _____ Due Date: 2026.09.30

This homework covers Lectures 6–7. There are three problems worth 20 points each. Show the main steps in each calculation and briefly explain the physical meaning of your results. You may use

$$\hbar = 1.055 \times 10^{-34} \text{ J s.}$$

Problem 1. Eigenstates and Measurement (20 points)

Let ϕ_1 and ϕ_2 be orthonormal eigenstates of a Hermitian operator $\hat{A}$:

$$\hat{A}\phi_1 = a_1\phi_1, \quad \hat{A}\phi_2 = a_2\phi_2, \quad a_1 \neq a_2.$$

The system is prepared in the state

$$\psi = \frac{\sqrt{3}}{2}\phi_1 + \frac{1}{2}\phi_2.$$

- (a) Verify that ψ is normalized. (4 points)
- (b) List the possible results of measuring A and give the probability of each result. (5 points)
- (c) Calculate the expectation value $\langle A \rangle$. (5 points)
- (d) Calculate $\hat{A}\psi$. Is ψ an eigenstate of $\hat{A}$? Explain why calculating $\hat{A}\psi$ is not the same as performing a measurement of A . (6 points)

Problem 2. The Position–Momentum Commutator (20 points)

In the position representation,

$$\hat{x}f(x) = xf(x), \quad \hat{p}f(x) = -i\hbar \frac{df}{dx}.$$

Let $f(x)$ be a differentiable test function.

- (a) Calculate $\hat{x}\hat{p}f(x)$ and $\hat{p}\hat{x}f(x)$. Use the product rule in the second calculation. (6 points)

- (b) Subtract the two results to show that

$$[\hat{x}, \hat{p}]f(x) = i\hbar f(x),$$

and hence state the operator identity for $[\hat{x}, \hat{p}]$. (5 points)

- (c) What does the nonzero commutator tell us about position and momentum as observables? (3 points)

- (d) A state has position uncertainty $\Delta x = 0.20 \text{ nm}$. Use

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

to find the minimum possible value of Δp . Give your answer in kg m/s . (6 points)

Problem 3. A Free-Particle Stationary State (20 points)

Consider a free particle on a periodic interval $0 \leq x \leq L$. In Lecture 7, the spatial part of a stationary state was denoted by $\phi_E(x)$, where $\hat{H}\phi_E = E\phi_E$. For a free particle it is convenient to label the same energy eigenfunction by its wave number k : we write $\phi_k(x) \equiv \phi_{E_k}(x)$. Let

$$\phi_k(x) = \frac{1}{\sqrt{L}} e^{ikx}, \quad k = \frac{2\pi n}{L}, \quad n \in \mathbb{Z},$$

and use

$$\hat{p} = -i\hbar \frac{\partial}{\partial x}, \quad \hat{H} = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}.$$

- (a) Verify that ϕ_k is normalized on $0 \leq x \leq L$, and show that it is a momentum eigenfunction. State its momentum eigenvalue. (5 points)
- (b) Show that ϕ_k is also an energy eigenfunction and find its energy eigenvalue E_k . (5 points)
- (c) If $\psi(x, 0) = \phi_k(x)$, write the complete time-dependent wave function $\psi(x, t)$. Verify directly that it satisfies the time-dependent Schrödinger equation. (5 points)
- (d) Calculate $|\psi(x, t)|^2$ and explain why this is called a stationary state. Compare $\phi_k(x)$ with $\phi_{-k}(x)$: what are their momenta and energies? (5 points)

End of Homework 3